



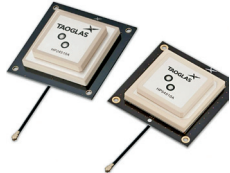
tracking. The Portenta UWB Shield, featuring Truesense DCU150, adds real-time location services and two-way ranging to Portenta projects, supporting industrial automation, logistics, and IoT applications with high precision and cloud connectivity.

Arduino

<https://www.arduino.cc>

Multi-band GPS antennas for precise positioning

Taoglas has introduced the Levity series, a line of active, multi-band GNSS antennas for positioning applications. The AHP24510 (L1/L2/L-band) and AHP54510 (L1/L5/L-band) directional patch antennas support



GPS, Galileo, GLONASS, and BeiDou satellite constellations, enabling signal acquisition in

various environments. With L-band capability, these antennas work with GNSS correction services to improve positioning accuracy to within 200cm by correcting for ionospheric delays and other errors. Multi-band acquisition also improves reliability and reduces energy use by shortening signal acquisition time. These antennas are used in wearables, navigation, transportation, robotics, precision agriculture, and autonomous vehicles.

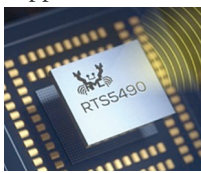
Taoglas

<https://www.taoglas.com>

World's first multi-port USB4 hub controller

Realtek Semiconductor Corp. has introduced the RTS5490, a multi-port USB4 hub controller with USB Type-C/PD functionality and USB-IF certification. The controller increases bandwidth from USB 3.2's 20Gbps to USB4's 40Gbps while supporting USB 3.2, USB-C, USB PD, PCIe, and DisplayPort. It remains compatible with USB 3.2 and USB 2.0 devices, allowing users to expand ports with a USB-C cable for data transfer, multi-display support,

and 240W charging. With USB4 support on downstream ports, the RTS5490 enables AI acceleration using USB4/Thunderbolt external graphics cards and data sharing



between computers through inter-domain transfer technology. It includes a 32-bit processor, USB 2.0/3.2 hub, DisplayPort controller, PCIe switch, and USB 3.2 host controller.

Realtek Semiconductor Corp.

<https://www.realtek.com>

Small, fast, and powerful car module

LG Innotek has launched the automotive application processor module (AP module) for the automotive semiconductor market. The AP module integrates



and controls vehicle electronic systems like advanced driver assistance systems (ADAS) and digital cockpits, functioning as the

vehicle's processor. With connected car advancements, including autonomous driving, demand for AP modules is increasing. Existing PCB-based semiconductor chips process large amounts of data from digital cockpits with ADAS and high-resolution displays. The AP module measures 6.35cm × 6.35cm (2.5 inches × 2.5 inches) and contains over 400 components, including a memory semiconductor, power management integrated circuit (PMIC), and system on chip (SoC).

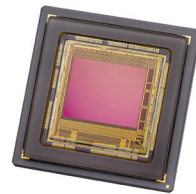
LG Innotek

<https://www.lginnotek.com/main/main.do>

High-speed sensor with extended sensitivity in NIR wavelengths

Teledyne e2v has introduced the Lince5M NIR, a CMOS image sensor designed for performance in visible and near-infrared (NIR) wavelengths. This sensor builds on Teledyne e2v's imaging technology to support

commercial, industrial, and medical applications. It is a monochrome sensor with a 5.2-megapixel resolution. It maintains the speed of the Lince5M while improving quantum efficiency across visible and NIR wavelengths.



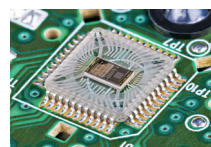
It delivers 250fps at full resolution using 24 LVDS output channels, making it suitable for applications requiring clear images in fast and low-light conditions. These include motion capture, sports analytics, industrial metrology, retinal imaging, and traffic monitoring. Built with Teledyne e2v's 5µm global shutter pixel, it offers a 55dB dynamic range in standard mode and over 100dB in high dynamic range mode for contrast-heavy scenes.

Teledyne e2v

<https://www.teledyne-e2v.com/en-us>

New generation of SPAD-based image sensors

Singular Photonics integrates computation into SPAD-based image sensing. This allows in-pixel and cross-pixel storage and processing at low light levels, revealing photon interactions. By capturing depth and temporal



data, Singular's sensors extract information from light, supporting applications in consumer

electronics, automotive technology, scientific research, and medical imaging. The company's 3D-stacked sensors perform photon counting, timing, and in-pixel statistical analysis. Sirona is a 512-pixel SPAD-based line sensor for time-correlated single-photon counting (TCSPC). It supports Raman spectroscopy, fluorescence lifetime imaging microscopy (FLIM), time-of-flight imaging, and quantum computing, with on-chip histogramming and time binning for spectroscopy.

Singular Photonics

<https://singularphotonics.com>